

2016.0 RANGE ROVER (LG), 303-08

ENGINE EMISSION CONTROL - TDV6 3.0L DIESEL - GEN 2/TDV6 3.0L DIESEL

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the engine emission control system and operation, refer to the relevant Description and Operation section of the workshop manual. REFER to: [Engine Emission Control](#) (303-08A Engine Emission Control - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Description and Operation).

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle.

NOTE:

Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Engine breather system ▪ Oil separator ▪ Exhaust Gas Recirculation (EGR) pipes/hoses (check for cracks) ▪ Exhaust Gas Recirculation (EGR) valve(s) ▪ Exhaust Gas Recirculation (EGR) cooler(s) ▪ Vacuum system	▪ Fuse(s) ▪ Wiring harness ▪ Loose or corroded electrical connector(s) ▪ Intake air shut off throttle ▪ Exhaust Gas Recirculation (EGR) valve(s) ▪ Engine Control Module (ECM)

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Difficult to start	▪ Exhaust Gas Recirculation (EGR) valve stuck open	▪ Using the manufacturer approved diagnostic system, perform routine - Inline diagnostic unit 2

Poor/erratic idle		diagnostic test - Exhaust gas recirculation valve. Clear the DTCs and retest
Lack of power when accelerating		
Engine stops/stalls	- Exhaust Gas Recirculation (EGR) valve stuck open - Breather system disconnected/restricted/blocked	- Using the manufacturer approved diagnostic system, perform routine - Inline diagnostic unit 2 diagnostic test - Exhaust gas recirculation valve. Clear the DTCs and retest - Check for Exhaust Gas Recirculation (EGR) DTCs - Check the engine breather system - Check the oil separator
Excessive fuel consumption	- Exhaust Gas Recirculation (EGR) valve stuck open - Exhaust Gas Recirculation (EGR) not operating - Breather system restricted/blocked	
Excessive black smoke		
Excessive emissions		
Excessive blow-by	Breather system restricted/blocked	- Check the engine breather hoses - Check the oil separator
Engine oil leaks	Breather system restricted/blocked	- Check the engine breather hoses - Check the oil separator

DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

REFER to: [Diagnostic Trouble Code Index - TDV6 3.0L Diesel - Gen 1.5, DTC: Engine Control Module \(ECM\)](#) (100-00 General Information, Description and Operation).